Advanced Machine Learning, ETH Zurich

The experimental studies

What happens, what counts, and what you decide

Carlos Cotrini

Autumn semester 2026

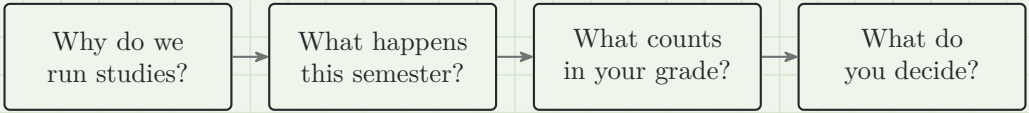
Subject

Four questions about the studies of this course

Entry

1

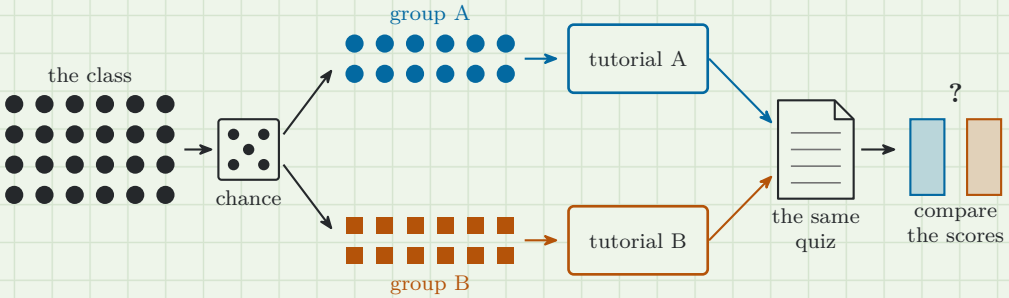
This semester, two topics of the course are taught in two versions, and two graded quizzes measure what you learned. This deck answers four questions about it, in this order.



Entry 1

Why we run studies

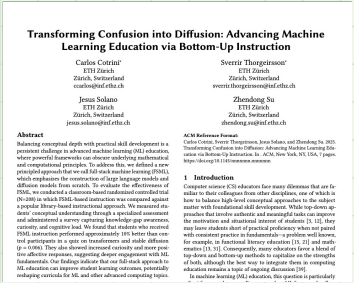
How do we find out whether one way of teaching a topic works better than another? We teach the topic both ways, each way to one group of the class drawn by chance, and give both groups the same quiz.



Chance decides the group of each student, as a coin toss would.
If one group scores higher on average, by more than chance would explain, the difference comes from its tutorial.

In autumn 2024, 208 students of this course learned transformers and diffusion models in one of two ways, drawn by chance, and a quiz measured what they understood.

- **Full stack:** you build every part of the model yourself before you reach for a library.
- **Library first:** you start from a library that already has the parts.
- The full stack group scored about **10 percent higher** on conceptual understanding ($p = 0.006$), and came out more curious.



Transforming Confusion into Diffusion: Advancing Machine Learning Education via Bottom-Up Instruction

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Abstract

Balancing conceptual depth with practical skill development is a persistent challenge in advanced machine learning (ML) education, where powerful frameworks can obscure underlying mathematical and computational principles. To address this, we define[a new, open-source curriculum that we use to teach machine learning (ML) which emphasizes the construction of a large language models and diffusion models from scratch. To evaluate the effectiveness of FSML, we conducted a classroom-based randomized controlled trial (N=208) in which FSML-based instruction was compared against a popular library-based instructional approach. We measured students' conceptual understanding through a specialized assessment and administered a survey capturing knowledge gap awareness, curiosity, and cognitive load. We found that students who received FSML instruction performed approximately 10% better than control participants in a quiz on transformers and stable diffusion ($p = 0.006$). They also showed increased curiosity and more positive affective responses, suggesting deeper engagement with ML fundamentals. Our findings indicate that our full-stack approach to ML education can improve student learning outcomes, potentially reshaping curricula for ML and other advanced computing topics.

ACM Reference Format:

Carlos Cotrini, Sverrir Thorgeirsson, Jesus Solano, and Zhendong Su. 2025. Transforming Confusion into Diffusion: Advancing Machine Learning Education via Bottom-Up Instruction. In ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

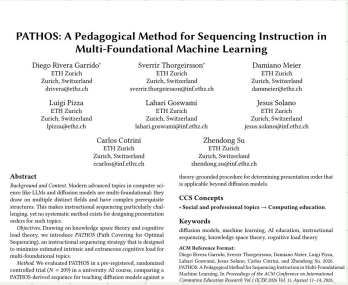
Computer science (CS) educators face many dilemmas that are familiar to their colleagues from other disciplines, one of which is how to balance high-level conceptual approaches to the subject matter with foundational skill development. While top-down approaches that involve authentic and meaningful tasks can improve the motivation and situational interest of students [3, 12], they may leave student short of practical proficiency when not paired with consistent practice in fundamentals—a problem well-known, for example, in functional literacy education [13, 21] and mathematics [15, 16]. Consequently, many educators favor a blend of both [4, 6, 11]. Nevertheless, striking the right balance—the strengths of both, although the best way to integrate them in computing education remains a topic of ongoing discussion [39].

In machine learning (ML) education, this question is particularly

Cotrini, Thorgeirsson, Solano and Su, SIGCSE TS 2026, **CER Best Paper (Global) award**.
doi:10.1145/3770762.3772595

In autumn 2025, 209 students of this course learned diffusion models from the same material, presented in one of two orders drawn by chance, and a quiz measured what they learned.

- **Order of discovery:** the ideas in the order in which the field found them.
- **Designed order:** an order built from what each idea needs first, since diffusion models draw on several fields at once.
- The designed order scored higher ($p = 0.011$, effect size $d = 0.32$), and its students reported a lower cognitive load.



PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning

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Abstract

Background and Context. Modern advanced topics in computer science like LLMs and diffusion models are multi-foundational: they draw on multiple distinct fields and have complex prerequisite structures. This makes instructional sequencing particularly challenging, yet no systematic method exists for designing presentation orders for such topics.

Objectives. Drawing on knowledge space theory and cognitive load theory, we introduce *PATHOS* (Path Covering for Optimal Sequencing), an instructional sequencing strategy that is designed to minimize estimated intrinsic and extraneous cognitive load for multi-foundational topics.

Method. We evaluated PATHOS in a pre-registered, randomized controlled trial ($N = 209$) in a university AI course, comparing a PATHOS-derived sequence for teaching diffusion models against a theory-grounded procedure for determining presentation order that is applicable beyond diffusion models.

CCS Concepts

• **Social and professional topics** → **Computing education.**

Keywords

diffusion models, machine learning, AI education, instructional sequencing, knowledge space theory, cognitive load theory

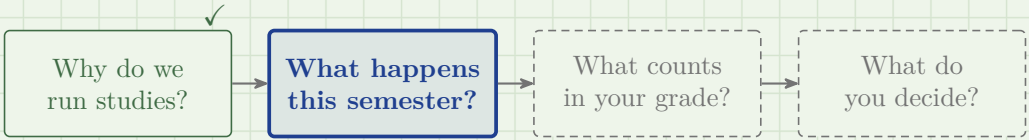
ACM Reference Format:

Diego Rivera Garrido, Sverrir Thorgeirsson, Damiano Meier, Luigi Pizza, Lahari Goswami, Jesus Solano, Carlos Cotrini, Zhendong Su. 2026. PATHOS: A Pedagogical Method for Sequencing Instruction in Multi-Foundational Machine Learning. In Proceedings of the ACM Conference on International Computing Education Research V.0 (CCSE 2026 Vol. 1). Aarust 11–14, 2026.

Rivera Garrido, Thorgeirsson, Meier, Pizza, Goswami, Solano, Cotrini and Su, ICER 2026.
doi:10.1145/3765964.3811657

Entry 2

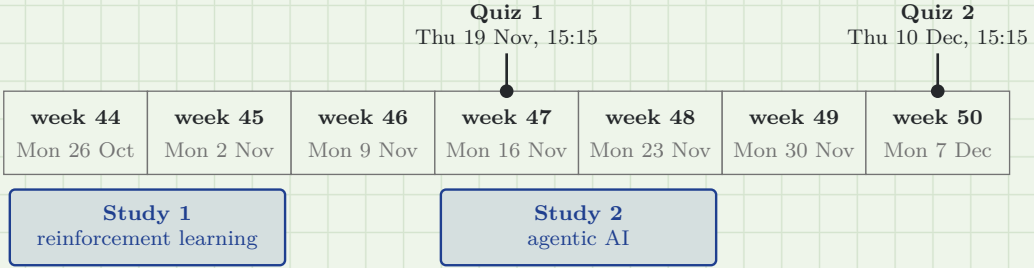
What happens this semester



So far. We compare two ways of teaching on two groups drawn by chance, and this course did so in 2024 and in 2025.

Now. The two studies of this semester: their topics, their weeks and their quizzes.

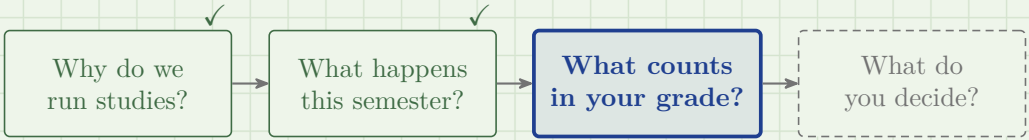
Study 1 is about reinforcement learning and study 2 about agentic AI. In each study, group A gets tutorial A and group B gets tutorial B for two weeks, and a quiz follows about two weeks later.



Study 1, reinforcement learning: the tutorials of weeks 44 and 45, from 28 October to 6 November.
Study 2, agentic AI: the tutorials of weeks 47 and 48, from 18 to 27 November.

Entry 3

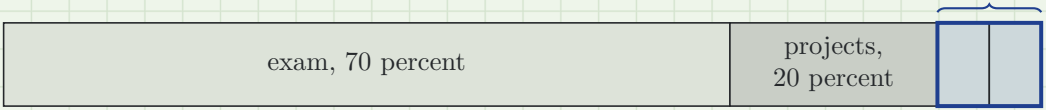
The quizzes and your grade



So far. Reinforcement learning in weeks 44 and 45 and agentic AI in weeks 47 and 48, each taught with tutorial A and tutorial B and followed by a quiz.

Now. How the two quizzes count for your grade.

Your final grade adds up the exam, the projects and the two quizzes with fixed weights, and it is rounded to the nearest quarter.



$$\text{final grade} = 0.70 \cdot \text{exam} + 0.20 \cdot \text{projects} + 0.10 \cdot \text{quizzes}$$

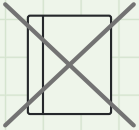
Both quizzes start at 15:15 on a Thursday. Both groups write the same quiz, and it is graded against the same fixed baselines for everyone.



45 minutes



on paper

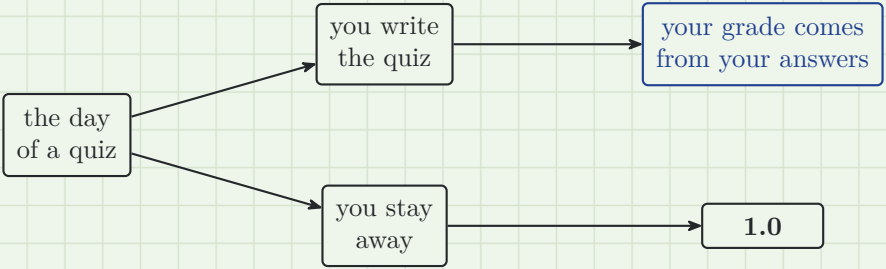


closed book

Quiz 1: Thursday 19 November, 15:15 to 16:00.
Quiz 2: Thursday 10 December, 15:15 to 16:00.

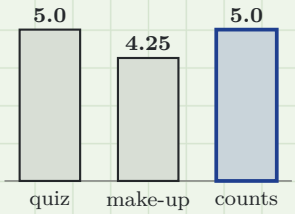
Rooms: we will tell you separately.
If you are entitled to a compensation for disadvantages, contact Carlos Cotrini, `ccarlos@inf.ethz.ch`.

Attending a quiz is voluntary, and every quiz is graded. A quiz you do not take counts as 1.0 in your final grade, unless the make-up raises it.

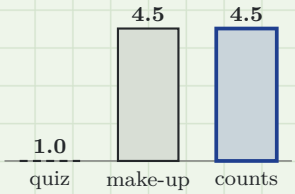


Two weeks after each quiz, a make-up quiz opens on Moodle. It is open book and lasts 45 minutes, you choose when to take it, and the better of your two grades counts.

You wrote the quiz



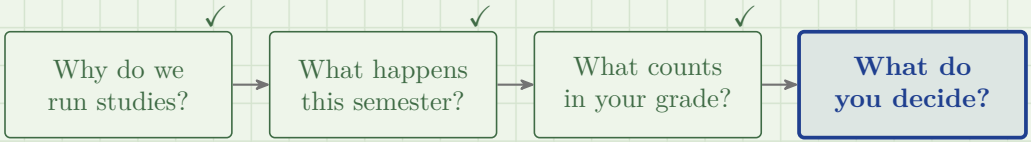
You missed the quiz



For the make-up, tutorial A and tutorial B are open to everyone.

Entry 4

Your consent and your data



So far. Each quiz counts 5 percent, attending is up to you, a missed quiz counts 1.0, and the make-up can only raise a grade.

Now. What you decide about the research use of your answers, and what happens to them.

The quiz belongs to the course, and it is graded the same way whatever you decide about the study. Joining the study means that the researchers may analyse your answers.

The quiz

part of the course
graded, 5 percent of your final grade
the same quiz for both groups

The study

may the researchers analyse your answers?
you decide: yes or no
your grade is the same either way

Your tutorials come before the quiz, so your decision changes nothing about them either.

With each quiz you get a consent form. On it you say whether the researchers may analyse your answers. If you say yes, it also asks your age, your gender and your education level.



your quiz

+

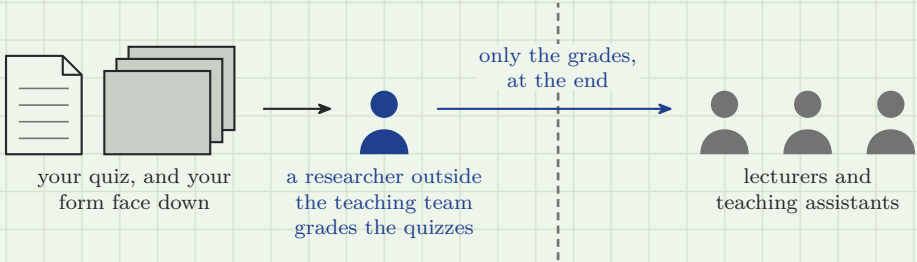
Consent form

☐ Yes, the researchers may analyse my answers.

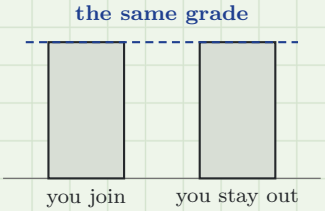
☐ No.

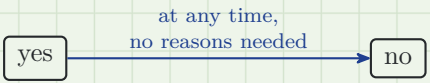
If yes: your age, your gender, your education level.

The consent forms are collected face down. Your quiz and your form go to a researcher outside the teaching team, who grades the quizzes. The lecturers and the teaching assistants get only the grades, at the end, and never see the forms.



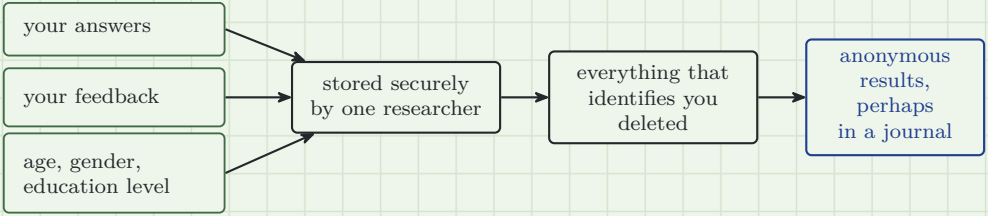
Joining the study is voluntary. You can withdraw at any time, without giving reasons and without any disadvantage, and your grade is the same whether you join or not.





Questions, or a change of mind: the contact person of the study is Sverrir Thorgeirsson, `sverrirt@ethz.ch`.

The researchers keep your quiz answers, your feedback on the quiz and, if you gave them, your age, gender and education level. After processing they delete everything that identifies you, and they may publish anonymous results.



The risks are minimal: a data breach could affect your privacy, and the researchers take every precaution to store your data safely.

The study was reviewed and approved by the ETH Zurich Ethics Commission under number 26 ETHICS-315. ETH's public liability insurance covers any damage related to it.

Questions about the study

Dr Sverrir Thorgeirsson
sverrirt@ethz.ch

Complaints

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Your data

Tomislav Mitar, data protection officer of ETH
tomislav.mitar@sl.ethz.ch

The information sheet of the study was sent to you by e-mail.

The study team: Dr Sverrir Thorgeirsson, Stefan Cadar, Sofi Mousia, Umut Demirbas, Johannes Schulte-Vels, Nikolaj Murasov, Jonathan Maillefaud, Oscar Jauffret. Insurance: Basler Versicherungen, policy no. 30/4.078.362.

Entry 5

Summary

Four steps, in the order of the semester.

1	**Weeks 44, 45, 47, 48**	Follow the tutorials of your group: reinforcement learning, then agentic AI.
2	**Thu 19 Nov and Thu 10 Dec, 15:15**	Write the two quizzes, 45 minutes each, 5 percent of your final grade each.
3	**On each quiz**	Decide whether the researchers may analyse your answers. Your grade is the same either way.
4	**Two weeks after each quiz**	Take the make-up on Moodle if you missed the quiz or want a better grade. The better grade counts.